

# GEOGRAPHY

# INDIAN GEOGRAPHY



## MINERALS & INDUSTRIES

### MODULE - 3

- GEOTHERMAL & TIDAL ENERGY
- COAL & ITS TYPES



# INDIAN MINERAL & ENERGY RESOURCES

## NON-CONVENTIONAL ENERGY SOURCES

### GEOHERMAL ENERGY



**1** When the magma from the interior of earth, comes out on the surface, tremendous heat is released.

**2** This heat energy can successfully be tapped and converted to electrical energy.

**3** The **hot water that gushes out through the geyser wells** is also used in the generation of thermal energy. It is popularly known as Geothermal energy.

**4** This energy is now considered to be one of the key energy sources which can be developed as an alternate source.

**5** The hot springs and geysers are being used since medieval period.



# INDIAN MINERAL & ENERGY RESOURCES

## NON-CONVENTIONAL ENERGY SOURCES

### GEOHERMAL ENERGY

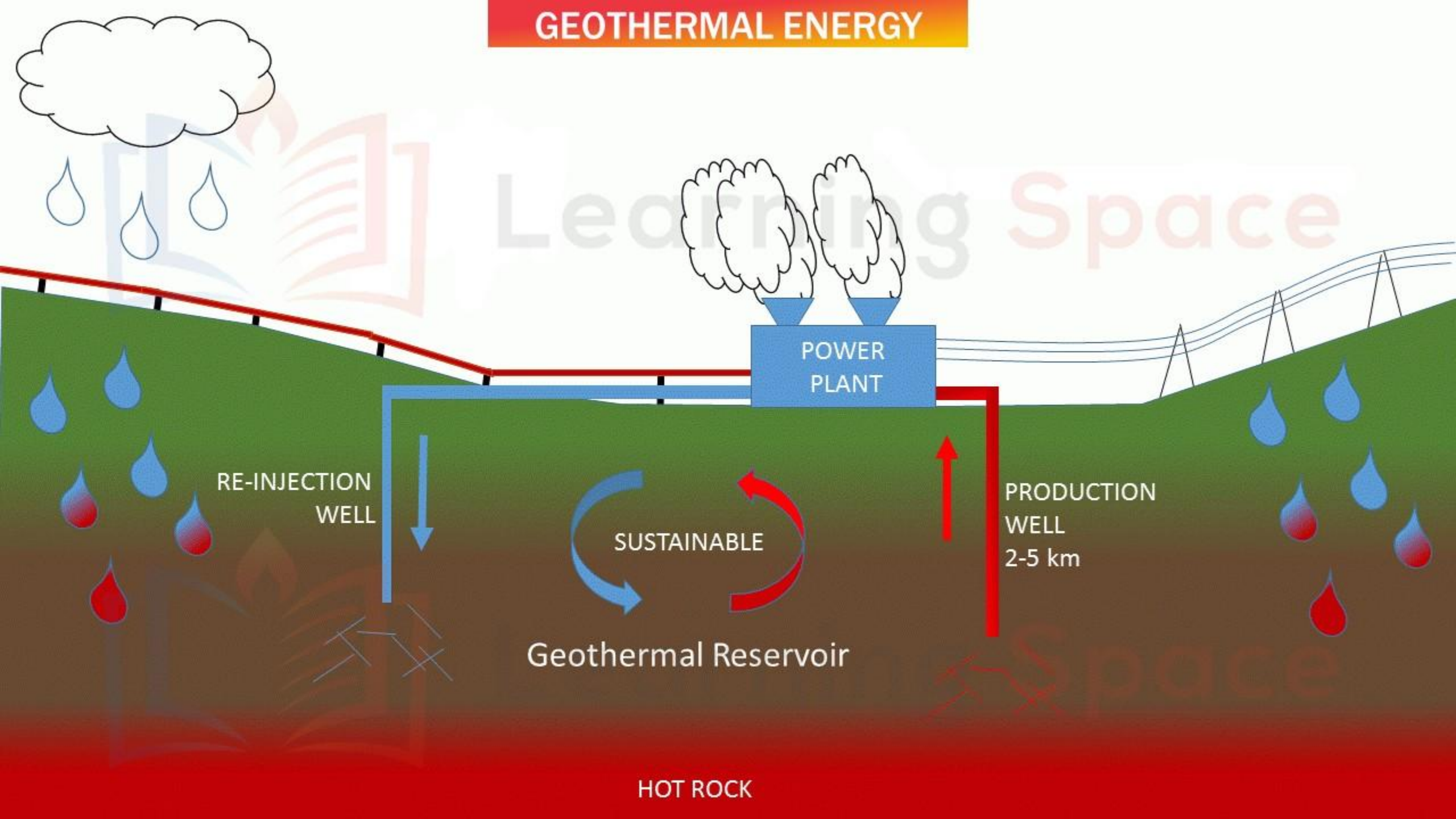


*The first successful (1890) attempt to tap the underground heat was made in the city of Boise, Idaho (U.S.A.), where a hot water pipe network was built to give heat to the surrounding buildings. This plant is still working.*

- 6 In some countries, such as in the USA water is pumped from underground hot water deposits and used for heating of houses.
- 7 Significant producers of this energy around the world are the Philippines, Iceland, New Zealand, United States.
- 8 The most significant capacity increases since 2004 were seen in Iceland and Turkey. Both countries doubled their capacity.
- 9 In India, a geothermal energy plant has been commissioned at Manikaran in Himachal Pradesh. Another potential site of geothermal energy is Puga valley in Kashmir.



# GEO THERMAL ENERGY





# INDIAN MINERAL & ENERGY RESOURCES

## NON-CONVENTIONAL ENERGY SOURCES

### GEOHERMAL ENERGY

5

Geothermal Energy (Other than Power) can also be applied in Tourist Resorts and pools, Melting snow, Poultry & Fish Farming, Horticulture, Greenhouses, Aquaculture, Industrial processes, Space heating, etc.

1

**Clean Energy Source:** It is a clean fuel as compared to fossil fuel such as oil, gas or coal. A geothermal field emits only  $\frac{1}{6}$ th  $\text{CO}_2$  as compared to any clean natural gas fired power plant.

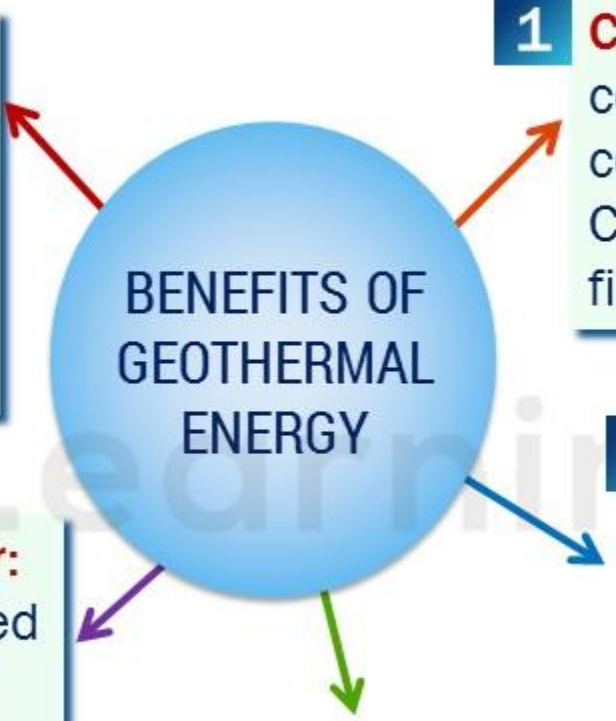
2

They are **naturally available Ground Source Heat Pumps**. It provides good heating solutions in colder climates.

3

The capital cost of geothermal power plant is very high but at the same time, **average units produced per year is very high** as compared to other renewable energy resources such as solar, hydro, wind, etc.

#### BENEFITS OF GEOHERMAL ENERGY



4

**Reinjection/Recycle of Ground Water:** Salts and dissolved minerals contained in geothermal fluids are **usually re-injected with excess water back into the reservoir** at a depth well below groundwater aquifers. This system prolongs the life of the reservoir as it **recycles the treated waste water**.



# INDIAN MINERAL & ENERGY RESOURCES

## GEOHERMAL ENERGY

### CASE STUDIES

#### PHILIPPINES

- 1 The Philippines is located in the **western part of the Pacific Rim** (part of the larger Ring of Fire).
- 2 The **volcanism** in the region occurs due to deep fractures in the earth, creating several geothermal centres.
- 3 The islands also experience a **healthy amount of rainfall** which recharges the groundwater and makes for an ideal environment to develop geothermal power plants.

#### ICELAND

- 1 Iceland sits right on top of the **mid-Atlantic ridge**, providing an ideal setting for **extensive volcanism**.
- 2 Due to its isolated location and the lack of other energy sources, Iceland has **invested heavily** in geothermal energy.
- 3 About 90% of Icelandic households directly use geothermal reservoirs for **heating purposes** and over one-fourth of the island's **electricity** is produced by five major geothermal power plants.



# INDIAN MINERAL & ENERGY RESOURCES

## GEOHERMAL ENERGY

### CASE STUDIES

#### HIMALAYAN GEOHERMAL BELT

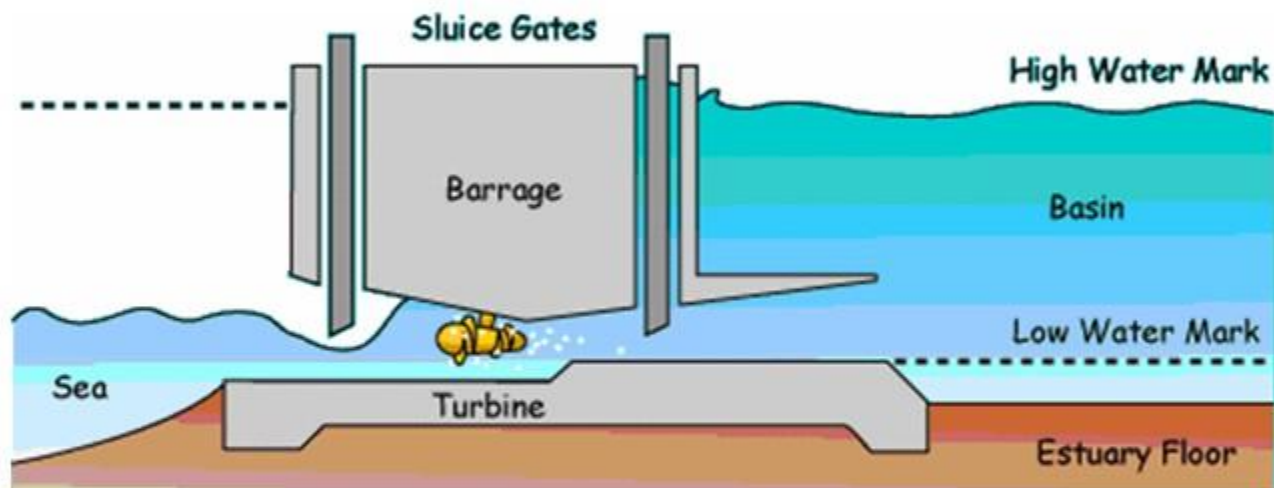
- 1 The HGB is a **large geothermally active area** that stretches across the **entire Himalayan orogeny** spanning half a dozen countries.
- 2 This region currently produces a **negligible amount of electricity** compared to its actual potential.
- 3 Many sites across India, China (Tibet), Nepal, Myanmar, and Thailand have been identified to be hot enough to produce electricity.
- 4 The geothermal activity is the result of the collision which happened between India and Eurasia around 40 million years ago.
- 5 The collision **causes the Indian Plate to subduct under Asia**, this caused the heating of large **granitic batholiths** under the Himalayas which heat the subsurface water.
- 6 The water is **recharged by large glaciers**, creating an ideal environment for geothermal energy exploitation.



# INDIAN MINERAL & ENERGY RESOURCES

## NON-CONVENTIONAL ENERGY SOURCES

### TIDAL AND WAVE ENERGY



- 1 Ocean currents are the store-house of infinite energy. For a very long time, efforts were made to create a more efficient energy system from the ceaseless tidal waves and ocean current.

- 2 Tidal power projects attempt to harness the energy of tides as they flow in and out.

- 3 The main criteria for a tidal power generation site is that for profitable energy generation the mean tidal range must be greater than 5 metres.

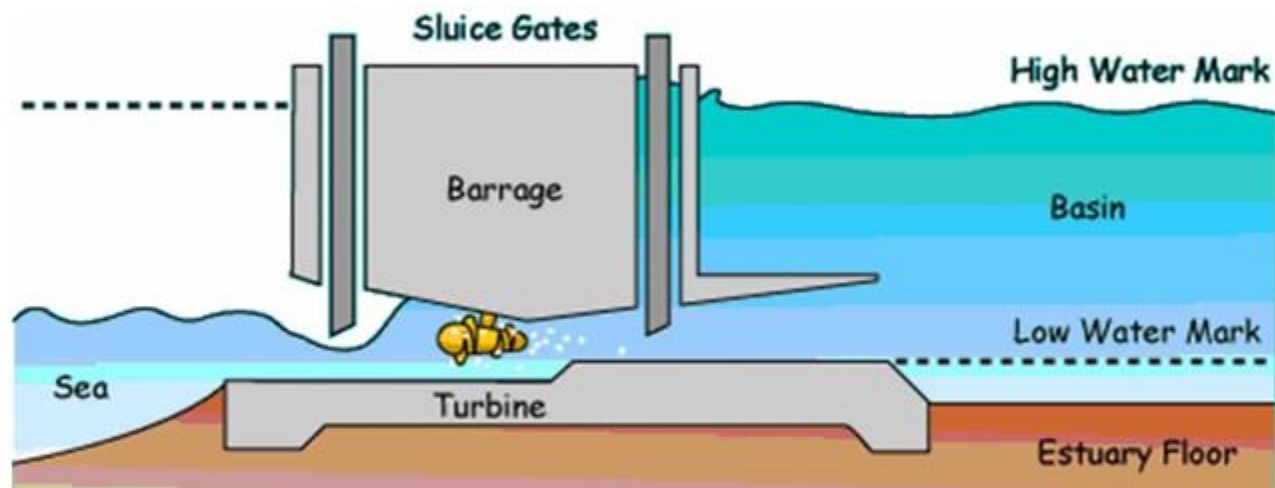
- 4 The tidal power is harnessed by building a dam across the entrance to a bay or estuary creating a reservoir.



# INDIAN MINERAL & ENERGY RESOURCES

## NON-CONVENTIONAL ENERGY SOURCES

### TIDAL AND WAVE ENERGY



- 5 As the tide rises, water is initially prevented from entering the bay. Then when tides are high and water is sufficient to run the turbines, the dam is opened and water flows through it into the reservoir (the bay), turning the blades of turbines and generating electricity.

- 6 Again when the reservoir (the bay) is filled, the dam is closed, stopping the flow and holding the water in reservoir. When the tide falls (ebb tide), the water level in the reservoir is higher than that in the ocean.

- 7 The dam is then opened to run the turbines (which are reversible), electricity is produced as the water is let out of the reservoir.

- 8 Large tidal waves are known to occur along the west coast of India.

- 9 Hence, India has great potential for the development of tidal energy.



## MINERAL FUELS

## COAL

1

- Coal is a one of the important minerals which is mainly used in the generation of **thermal power and smelting of iron ore**.
- Coal occurs in rock sequences mainly of two geological ages, namely **Gondwana and Tertiary deposits**.

### GONDWANA COAL

- Belongs to the **Carboniferous period** around 300 million years ago.
- Found in **Damodar, Mahanadi, Godavari valleys** in the Peninsular plateau.
- Consist of mainly **bituminous or anthracite coal** varieties.
- Majority of coal reserves in India belongs to this type.

### TERTIARY COAL

- Found in the **Oligocene period** of the Tertiary Era around 40 million years ago.
- Found in **Assam, Arunachal Pradesh, Meghalaya, Nagaland etc.**
- Consist of **inferior variety** of coal like lignite.



## MINERAL FUELS

## COAL

### GONDWANA COAL

**2** About 80 per cent of the coal deposits in India is of **bituminous** type and is of non-coking grade.

- 3** **Damodar Valley Coal Fields:**
- The most important Gondwana coalfields of India are located in Damodar Valley.
  - They lie in **Jharkhand-Bengal coal belt** and the important coal fields in this region are **Raniganj, Jharia, Bokaro, Karanpura**.
  - Jharia is the largest coal field followed by Raniganj.

**4** The other river valleys associated with coal are Godavari, Mahanadi and Sone.

- 5** The most important coal mining centres are :
- **Singrauli** in Madhya Pradesh (part of Singrauli coal field lies in Uttar Pradesh)
  - **Korba** in Chhattisgarh
  - **Talcher** and Rampur in Odisha
  - Wardha, Kamptee and Bander in Maharashtra
  - **Singareni** in Telangana
  - Pandur in Andhra Pradesh.



## MINERAL FUELS

## COAL

### TERTIARY COAL



- 6 Tertiary coals occur in Assam, Arunachal Pradesh, Meghalaya and Nagaland.
- 7 It is extracted from Darangiri, Cherrapunji, Mewlong and Langrin (Meghalaya); Makum, Jaipur and Nazira in upper Assam, Namchik – Namphuk (Arunachal Pradesh) and Kalakot (Jammu and Kashmir).
- 8 Besides, the brown coal or lignite occur in the coastal areas of Tamil Nadu, Pondicherry, Gujarat and Jammu and Kashmir.



## MINERAL FUELS

## COAL

### AT SINGARENI, CANARIES TO MINERS' RESCUE

## DYNAMIC ISSUES



- 9 Singareni collieries in Kothagudem , the country's premier coal production company used canaries to detect the presence of deadly carbon monoxide in underground mines.

- 10 Miners collapse and often die even if small quantities of the highly poisonous CO are present in the air.

- 11 Though, miners speak lovingly of the canaries, the underground experience is not at all pleasant for the birds.

- 12 When lowered into mines with CO presence, the birds show distress symptoms such as ruffling of feathers, pronounced chirping and loss of life.



## DYNAMIC ISSUES

AT SINGARENI, CANARIES TO  
MINERS' RESCUE

**13** These reactions occur even if **0.15 per cent of CO** is present in the air.

**14** If the content is 0.3 per cent the bird shows immediate distress and falls off its perch in two to three minutes

**15** The rapid breathing rate, small size and high metabolism of the canaries make them die before miners upon inhaling toxic gas.

**16** With the Singareni collieries introducing **state of the art hand held gas detectors** with sensors and alarm systems, the birds are not used much as an early warning system.

**17** The sophisticated hand held CO detectors introduced by the company can detect CO concentrations from as low as 10 ppm to as high as 1,000 ppm.



## A tragedy that was long in the making

Illegal rat-hole mining in Meghalaya persists despite ruinous effects on the environment



PATRICIA MUKHERJEE

The efforts to reach the 15 miners trapped in an illegal coal mine in the East Jaintia hills of Meghalaya since December 13 continue, but they began belatedly and have faced many problems.

### Doomed from the beginning

First, the Meghalaya government has no idea what happens inside these rat-hole mines, which are barely 2 ft wide, since mining is a private activity. Despite the National Green Tribunal ban of April 2014, mining continues in the State. Second, it was unfortunate that the district administration assumed the miners to be dead on the very day of the tragedy. This assumption was evident in the letter written to the National Disaster Response Force. It was only after a Delhi-based lawyer, Aditya N. Prasad, represented by senior Supreme Court advocate Anand Grover and his team of human rights lawyers presented their suggestions to the court that the Meghalaya government got different actors to the accident site. Mr. Prasad has never visited Meghalaya. When asked why he is the petitioner on behalf of the miners, he simply said: "They are fellow Indians and my brethren." That someone based in Delhi should have the empathy lacking in the people and the government shows that humanity is a dying virtue.

Mr. Prasad has done everything

possible to put things together to assist the rescue mission. But despite his initiative, things were delayed. The distance of the mine, for one, was a major hindrance. Then there are other issues that need to be highlighted. The trapped miners were being racially profiled in the minds of the people and the state. Of the 15 miners, only three were locals from the nearby village of Lumthari. The rest were Muslims from Garo Hills, Meghalaya, and Bodoland, Assam. Their socio-economic profile also worked against them. They were the poorest of the poor who took a huge risk to enter a mine and dig for coal without any safety gear.

When a mine is flooded, the immediate response, apart from pumping out the water, is to stop further flow of water into it. This requires a hydrologist to scientifically map out the area from where water entered the mine. Sudhir Kumar, a hydrologist from the National Institute of Hydrology, Roorki, arrived only two weeks after the disaster. So did the divers from the Indian Navy and the 100 HP water pumps from Kiclodkar Brothers. The remotely operated underwater vehicle (ROV) from Planys in Chennai came three weeks later. So did the geologists from Hyderabad. All these delays happened because there was no one person or agency to coordinate the rescue mission. This shows the kind of disaster preparedness we have in our country. One shudders to think what the response would be if there was a massive earthquake in the Northeast, which is listed as Zone 5 on the seismic scale.

There are many questions that arise with respect to rat-hole min-



ing of coal. One, why does the state allow this archaic mining system, which has complete disregard for human life and safety? And two, why is Meghalaya exempted from national mining laws? Rat-hole mining, which started with gusto in the 1980s, has poisoned three rivers in the Jaintia hills: the Myntdu, Lunar and Lakha. Scientists from the North Eastern Hill University have found that these rivers have very high acidic levels. Reports from other agencies suggest that pH of the water and sulphate and iron concentrations indicate significant deterioration of the rivers. Acid mine drainage from abandoned mines was a major cause for water pollution in the areas investigated, the reports added.

The coal mine owners have been hiring the best legal brains to argue for them in the highest court of the land. They say that rat-hole mining should continue because no other form of mining is viable (which means that their profit margins would reduce if other forms of mining were to take place). They argue that the NCT ban should be lifted. They claim

that coal mining provides livelihoods for many, but at what cost?

### The fault-lines

The tribes of Meghalaya are divided on the issue of rat-hole mining. The fault-lines are clear. Those who care for the environment and for a future for their children and grandchildren have been clamouring for an end to the practice of rat-hole mining and reckless limestone mining. On the other hand, the mining elite have mobilised forces to demolish environmental activists. A community of just over a million is now fragmented. To add to these woes, cement companies also release their effluents into the rivers. So we now have a deadly cocktail of pollutants being released into the environment. The scale of the problem is clear in this one fact: there are 3,923 coal mines in one district with a geographical area of 2,226 sq. km.

The other troubling factor is that coal mine owners are insisting that since Meghalaya is a State under the Sixth Schedule of the Constitution, national mining laws should be exempted here. The Sixth Schedule was enacted to

protect the community rights of tribals from any form of exploitation of their land and resources. How can it now be used as an instrument to protect an activity that is a private enterprise, that is inhuman, and that violates Article 21 of the Constitution? Why is the Sixth Schedule unable to protect the forests and rivers that are common property resources? Acid mine drainage has rendered even agricultural land non-productive. Mine owners do not care about environmental degradation.

### Abandoning responsibility

The cement giant, Lafarge, mines limestone from Meghalaya's East Khasi Hills district and transports it to Chhatak in Bangladesh via conveyor belts. The Supreme Court placed a heavy penalty on Lafarge and asked it to strictly implement environmental laws apart from generating livelihoods for people residing within 50 km of the mining areas. In the case of coal mine owners, there are no such strictures. They have left thousands of abandoned mines as human graves. The State does not insist that they reclaim and afforest those mines. In 40 years of mining and profiteering, the mine owners have till date not constructed a single hospital or even a school. There is complete disregard for corporate social responsibility because the mines are privately owned by the tribals. How long can the Central government and the highest court of the land allow this to carry on in one part of the country when strict laws are applied elsewhere?

Patricia Mukherjee is Editor, The Shillong Times

## DYNAMIC ISSUES RAT HOLE MINING

### IMPORTANT QUESTION

### ESSAY/ MAINS / GD / INTERVIEWS

In addition to the loss of lives, Acid Mine Drainage is the major environmental concern in Meghalaya due to rat-hole mining. Discuss.



## Where coal is mined via 'rat-holes'

Despite a ban, rat-hole mining remains a prevalent practice in Meghalaya, where a mine has collapsed and 15 workers are feared dead. What makes this method unsafe, and why is it widely followed?

ABHISHEK SAHA  
GUWAHATI, DECEMBER 19

LAST WEEK, the collapse of a coal mine in Meghalaya's East Jaintia Hills, trapping at least 15 workers who were still missing until Wednesday and are feared dead, has thrown the spotlight on a procedure known as 'rat-hole mining'. Although banned, it remains the prevalent procedure for coal mining in Meghalaya. A look at how rat-hole mining is carried out, and why it is dangerous.

### What is rat-hole mining?

It involves digging of very small tunnels, usually only 3-4 feet high, which workers (often children) enter and extract coal. Of Singh, professor of environmental studies at North Eastern Hill University (NEHU) in Shillong, said the Indian Express that rat-hole mining is basically of two types. 'In side-cutting procedure, narrow tunnels are dug into the hill slopes and workers go inside until they find the coal seam. The coal seam in hills of Meghalaya is very thin, less than 2 m in most cases,' he said. In the other type of rat-hole mining, called box-cutting, a rectangular opening is made, varying from 10 to 100 sq m, and through that is dug a vertical pit, 100 to 400 feet deep. Once the coal seam is found, rat-hole-sized tunnels are dug horizontally through which workers can extract the coal.

### When was it banned, and why?

The National Green Tribunal (NGT) banned it in 2014, and retained the ban in 2015 on grounds of it being unscientific and unsafe for workers. The state government has appealed the order in the Supreme Court.

**ECOLOGIC:** In their petition to the NGT, Assam's All District Students' Union and the Dima Hasao District Committee complained that rat-hole mining in Meghalaya had caused the water in the Kopli river (it flows through Meghalaya and Assam) to turn acidic. The NGT order quoted a report by Prof Singh: 'Tunnel roadblocks in and around mining areas are used for piling of coal which is a major source of air, water and soil pollution. Off road movement of trucks and other vehicles in the area causes further damage to the ecology of the area.'

**RISK TO LIVES:** The NGT observed, 'It is also observed that there are a number of cases where by virtue of rat-hole mining, during the rainy season, water flooded into the mining areas resulting in death of many individuals including employees/workers.' Following the state's appeal, civil society groups this month submitted a 'Citizens Report' to Supreme Court-appointed anti-



NDRI personnel search for the missing workers in East Jaintia Hills, Meghalaya. PTI

corrupt Colloquialism: 'A worker carries with him a pickaxe, a shovel and a wheelbarrow. As the cave is dark he carries a torch. If water has seeped into the cave, the worker carries only after the water is pumped out. Workers usually enter the cave early in the morning and keep on working till they are tired or if they are hungry or when they feel that they have earned enough money for the day.'

### To what extent is the practice followed?

According to available government data, when the NGT ban was ordered, Meghalaya's annual coal production was nearly 6 million tonnes. Almost all of it is said to have come through rat-hole mining. Prof Singh explained that no other method would be economically viable in Meghalaya, where

mining but all 'unscientific and illegal mining'. The state has in place the Meghalaya Mines and Minerals Policy 2012, while clearances and permissions are required under central laws including the Mines and Minerals (Regulation & Development) Act. When the Meghalaya police arrested the main accused in the mine collapse case, he was charged under sections including MMRD Act section 21(1), which deals with penalties for mining without permission.

Highlighting the extent of illegal mining, the Governor's report to the SC-appointed anti-corruption court said: 'The 6th Schedule of the Constitution intends to protect the community's ownership over its land and the community's autonomy and consent over its natural resources. Coal mining currently underway in Meghalaya was a corruption of this Constitutional Provision wherein private individuals having private interests in earning monetary benefits from minerals vested under the land are engaging in coal mining and are attempting to legitimise this act by claiming autonomy through its local autonomy over land ownership, whereas the truth is that the land belongs to the community and not even to the individual.'

### Has the NGT ban not helped check this?

Following the mine collapse, Chief Minister Conrad Sangma said in a video that illegal mining does happen, and promised appropriate action. Agnes Khushim, a leading campaigner against illegal mining, and her colleague Anika Sangma were arrested in East Jaintia Hills last month while trying to locate an illegal mine. 'Activists speaking out have been threatened or attacked. Political elites are part of the scene and this mafia is very powerful and dangerous,' activist Angira Rongai, member of the group Tharu U Kangli Juti, told the Indian Express.

In its 2015 order, the NGT observed, 'It is indisputable that orders of the Tribunal have been violated without exception... The State Government has failed to check illegal mining effectively and has also not framed the mining policy, mining plan and the guidelines as directed under the orders of the Tribunal.'

### But does Meghalaya not have a policy?

The NGT finds the 2012 policy inadequate. The policy does not address rat-hole mining and central states: 'Small and traditional systems of mining by local people in their own land should be recognised and regulated by any law and extraction of coal has been made by unscientific elements in a most illegal and unscientific manner.' 'The State of Meghalaya has promulgated a mining policy of 2012, which does not deal with rat-hole mining, but on the contrary deprecates it.'

### When is mining illegal?

The NGT order bans not only rat-hole

## DYNAMIC ISSUES RAT HOLE MINING

### UPSC CSE – GS II

## GS II ... Social Justice and Governance

Government policies and interventions for development in various sectors and issues arising out of their design and implementation.



The Indian **EXPRESS**  
JOURNALISM OF COURAGE

20<sup>th</sup> December, 2018





## DYNAMIC ISSUES RAT HOLE MINING

### WHY PEOPLE RESORT TO RAT-HOLE MINING?



*Meghalaya's annual coal production was nearly 6 million tonnes, when the NGT ban was ordered.*

- 1 Almost all of the coal comes from rat-hole mining.
- 2 No other method would be economically viable in Meghalaya, where the coal seam is extremely thin.
- 3 Removal of rocks from the hilly terrain and putting up pillars inside the mine to prevent collapse would be costlier.
- 4 In Jharkhand, the coal layer is extremely thick. One can do open-cast mining, but, in Meghalaya this is locally developed technique and the most commonly used one.



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